

Sync Sofa: Sofa-type Side-by-side Communication Experience Based on Multimodal Expression

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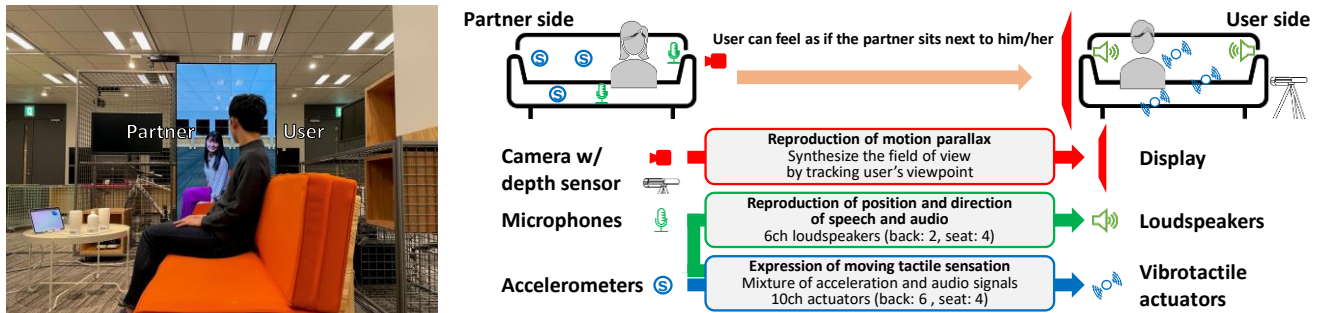


Figure 1: The picture of demonstration and system overview

ABSTRACT

Lifestyle changes and digitalization have reduced opportunities for face-to-face, intimate communication, which is an indispensable activity for human beings. We have realized a method that allows intimate communication between two persons even though they are faraway from each other by integrating multimodal technologies. Sync Sofa is a new sofa-type communication tool. It senses the partner with a camera, microphones, and accelerometers. The sensed data are cross-modally integrated and then presented to the user through a life-size display, multichannel loudspeakers, and multichannel vibrotactile actuators. We have received positive feedback from many users who have experienced Sync Sofa, such as, "I felt like the other person was really sitting next to me".

CCS CONCEPTS

- Applied computing ~ Arts and humanities ~ Media arts
- Hardware ~ Communication hardware, interfaces and storage ~ Tactile and hand-based interfaces ~ Haptic devices

KEYWORDS

Tactile, Free-viewpoint video, Audio rendering, Media arts

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1 Introduction

Due to the spread of Covid-19 and the increase in the number of single-person households, the opportunity for intimate contact and communication, such as touching an arm or shoulder, snuggling up and so on, are decreasing. The lack of such communication may have a negative impact on mental health, self-esteem, and the immune system [1][2][3][4]. One solution to this challenge is to realize such communication online.

We propose Sync Sofa, a new sofa-type communication system that gives users with the sense that their partner is sitting next to them, even when they are faraway. In terms of personal space [5], the situation of being side by side is considered to play an important role in achieving intimate communication without feeling discomfort. When you communicate with someone within touching distance, sitting side by side is more conducive to intimacy than sitting opposite because personal space tends to be established farther out directly in front of the body [5][6]. In addition, being side by side makes it easier to achieve a situation where the user can easily touch the partner's shoulders and back, which are generally touched in touch communication. Therefore, Sync Sofa has the

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potential to be an ideal platform for intimate communication between distant partners.

We realized Sync Sofa by integrating multimodal (visual/ audio/ tactile) expressions. The system allows users to have the experience that their partner is sitting next to them through the three modalities. The modal representations include natural motion parallax, sound localization highly linked to the partner's position and multi vibrotactile sensation presentation on the back and seat of the sofa.

2 System

The 3D design of Sync Sofa is shown in Figure 2. The figure focuses on the user side. The main difference between the user side and the partner side is that the camera is located in place of the display for capturing the partner. The technical elements of each of the visual, audio and tactile modalities are described below.

2.1 Visual expression

A 65-inch 4K monitor, placed next to where the user sits, shows an image of the partner, which is expressed as a billboard model [7]. Motion parallax was implemented to enhance the sense that the partner was present. To generate the parallax effect, a depth sensor, Azure Kinect [8], was installed in front of the sofa to track the user's head position. The billboard-based object corresponding to the partner is placed with background objects such as sofa and room in virtual 3D space, and then this system renders the virtual camera view synchronized with the tracked user's head position. Finally, the user can approach or look around the partner on the monitor without discomfort.

2.2 Audio expression

To balance the wide sound image localization and the simplicity of the system, six full-range loudspeakers are installed, two on either side of the back and four on either side of the seat. They are placed at a sufficient distance from each other so that the sound image can be widely localized by controlling the intensity of each loudspeaker or filtering the audio stream. In order to output motion sounds (the sounds of footsteps, rubbing, etc.) and speech sounds separately, different types of microphones are placed to capture motion sounds and speech sounds, respectively. The intensity of each loudspeaker and the filter for the directional band were parametrically and heuristically adjusted. The motion sounds were heard from the partner's sitting position and the speech sounds from the partner's face position. Finally, the user can hear the motion sounds and the speech sounds whose sound image localization is linked to the image on the monitor [9].

2.3 Tactile expression

Ten vibrotactile actuators are installed in the sofa, six on the back and four on the seat. The shoulder width and the buttock width of a typical Japanese person were used as references for designing the actuator positions. Each actuator is driven by an analog signal and provides a maximum drive power of 15W.

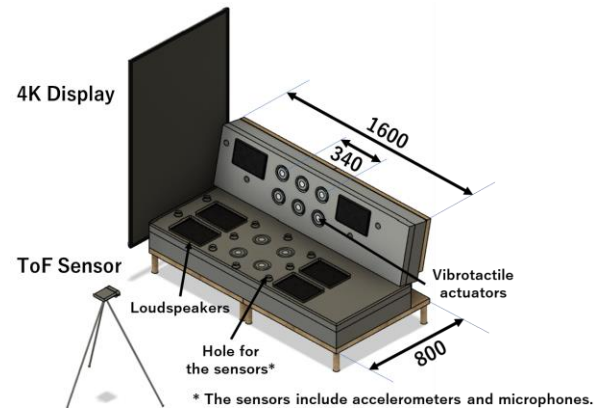


Figure 2: 3D Design

The drive signal is a mixture of an acceleration signal and an audio signal, and each sensor is placed on the back and the seat of the sofa the partner sitting on. By driving tactile vibrations from the acceleration signals, the user can experience all of the movements the partner performs precisely because the accelerometer has high spatial resolution. However, if only acceleration signals are used, the vibration becomes largely disengaged from the audio content, e.g., the motion sound is very small but the tactile vibration is very large.

Hence, we decided to use audio signals, too. This allows the user to feel tactile sensations that are highly linked not only to the partner's position and movement but also with the audio experience.

We constructed the system as shown in the configuration diagram (Figure 1) and realized the transmission of the sense that the partner is sitting next to the user in the local network.

3 Demonstration and future work

We performed an internal demonstration for about 100 men/women aged from their 20s to 60s, and we got positive feedback from most of them. In our demonstration, participants can experience Sync Sofa with pre-recorded content and a real-time communication demonstration. The four recorded scenes referred to below are prepared for the demonstration although just two representative scenarios are shown in the attached video. Each scenario has been created as content suitable for experiencing the functional features of Sync Sofa.

1. Scenario where the partner comes in and sits next to you.
2. Scenario where the partner rises from the sofa, sits down again and leans against the back.
3. Scenario where the partner comes about 20 cm closer to you on the sofa, strokes your back and then taps your back lightly.
4. Scenario where the partner is watching a soccer game on TV and cheers the moment a goal is scored.

The current system is realized as a one-way, peer-to-peer, and local network communication system. However, we plan to realize bidirectional, multi-location communication over an IP network in the future.

REFERENCES

- [1] Morrison, I. 2016. Keep Calm and Cuddle on: Social Touch as a Stress Buffer. *Adaptive Human Behavior and Physiology* 2, 344–362, DOI: <https://doi.org/10.1007/s40750-016-0052-x>.
- [2] Cohen, S., Janicki-Deverts, D., Turner, R. B., and Doyle, W. J. 2015. Does hugging provide stress-buffering social support? A study of susceptibility to upper respiratory infection and illness. *Psychological Science* 26(2), 135–147, DOI: <https://doi.org/10.1177/0956797614559284>.
- [3] Jakubiak, B., Feeney, B. 2016. Keep in touch: The effects of imagined touch support on stress and exploration. *Journal of Experimental Social Psychology* 65, 59–67, DOI: <https://doi.org/10.1016/j.jesp.2016.04.001>.
- [4] Thompson, E. and Hampton, J. 2011. The effect of relationship status on communicating emotions through touch. *Cognition & Emotion* 25(2), 295–306, DOI: <https://doi.org/10.1080/02699931.2010.492957>.
- [5] Sommer, R. 2008. *Personal Space: The Behavioral Basis of Design*, Bosko Books.
- [6] Alec, H. 1981. The shape of personal space: An experimental investigation. *Canadian Journal of Behavioral Science*, 13(1), 087–093. DOI:
- [7] Sankoh, H., Naito, S., Nonaka, K., Sabirin, H. and Chen, J. 2018. Robust Billboard-based, Free-viewpoint Video Synthesis Algorithm to Overcome Occlusions under Challenging Outdoor Sport Scenes. *Proc. 26th ACM International Conference on Multimedia*, 1724–1732, DOI: <https://doi.org/10.1145/3240508.3240514>.
- [8] Microsoft corporation. Azure Kinect DK documentation. <https://docs.microsoft.com/en-us/azure/kinect-dk/>
- [9] Horiuchi, T., Sankoh, H., Kato, T., and Naito, S. 2012. Interactive Music Video Application for Smartphones Based on Free-viewpoint Video and Audio Rendering. *Proc. 20th ACM International Conference on Multimedia*, 1293–1294, DOI: <https://doi.org/10.1145/2393347.2396449>.